

Chapter 4

Bayesian Model Checking

4.1 Model comparison versus model checking

To identify an appropriate model for a given dataset, we can employ two strategies.

1. Formal model comparison: This strategy involves listing all candidate models and assigning a prior probability to each. We then compute their posterior probabilities to either select the single model with the highest posterior probability, or using a weighted average of all candidates as described in Section 2.8.
2. Posterior predictive checks: This strategy evaluates models individually by determining if it is plausible that a model could generate data similar to what was observed in practice. By comparing the observed data against the model's simulated reality, we can identify specific areas where the model fails to capture the data-generating process, and improve the model by fixing them.

The model comparison approach is highly structured and fits neatly within the standard inferential framework. However, it only provides a relative ranking, and it does not tell you if the winning model is actually a good fit for the data. It offers no diagnostics regarding model failure nor any guidance on how to improve the existing candidates.

Unlike formal comparison, predictive checks act as a diagnostic tool. They provide the necessary hints to uncover what is wrong with a model, allowing for an iterative process of model criticism and improvement. This ensures that the final model is not just the best among a set of alternatives, but is good enough for the task at hand.

4.2 Model checking through predictive checks

Bayesian models are, at their core, data simulators. Therefore, the most natural way to assess a model is to evaluate the quality of the synthetic data it generates. By checking how well a model replicates the observed reality, we can identify its specific shortcomings and discover how to improve it.

The goal is to compare replicated data sets ($y^{rep} = (y_1^{rep}, \dots, y_n^{rep})$), simulated from the model's predictive distribution, with the real observed data ($y = (y_1, \dots, y_n)$). We ask ourselves whether the simulated data and the real data are indistinguishable or not.

Because real-world systems are complex, a model might succeed in some areas but fail in others. Therefore, we must focus our comparison on the specific features of the data, that are most relevant to our scientific goals. In practice, that means comparing data sets based on their location, spread, skewness, frequency of extreme values, or other more problem specific features.

While less formally structured than model comparison, predictive checking follows a consistent three-step process:

1. Choose specific statistics $T(y)$ or graphical displays that best summarize the data features of interest.
2. Select the appropriate predictive distribution (typically the posterior predictive distribution) to generate multiple sets of replicated data.
3. Determine the degree of conflict between the observed $T(y)$ and the distribution of replicated statistics $T(y^{rep})$.

In practice, this means we compare our single observed data set against a large ensemble of replicated data sets. If the observed data looks like a typical draw from the model's simulations, the model is plausible. If the observed data is an extreme outlier, the model is failing to capture a fundamental feature of reality.

4.2.1 Choice of statistics or graphics

The comparison between simulated and observed data should always prioritize the features of reality most relevant to the researcher. While the best approach is to tailor

summary statistics or graphics to the specific problem, several generic statistics are widely applicable for model diagnostic.

1. Density-based statistics, to assess how surprising each observation y_i is under the model. The most used ones are $p(y_i)$, $p(y_i|y)$ or the conditional predictive ordinate, $CPO_i = p(y_i|y_{(i)})$, which is the posterior predictive for $\tilde{y}_i$ conditioning on all the observations except y_i . A larger value indicates that the observation is well-supported by the model, while a very small value identifies y_i as an outlier.
2. Distance-based statistics, especially useful in the context of linear models. The most used ones are $(y_i - E(\tilde{y}_i))^2$, $(y_i - E(\tilde{y}_i|y))^2$ or $(y_i - E(\tilde{y}_i|y_{(i)}))^2$. These measure the distance between the observed data and a point prediction by the model. A smaller value indicates a better fit for the i -th observation.

To evaluate the model's performance across the entire data set, we can aggregate these point-wise diagnostics into global summary statistics:

1. Summing the logarithms of the densities, such as $\sum_i \log p(y_i|y)$ or $\sum_i \log CPO_i = \sum_i \log p(y_i|y_{(i)})$, or
2. Summing the squared distances, such as $\sum_i (y_i - E(\tilde{y}_i|y))^2$ or $\sum_i (y_i - E(\tilde{y}_i|y_{(i)}))^2$.

In both cases, we use these sums as test statistics $T(y)$. By comparing the observed $T(y)$ to the distribution of $T(y^{rep})$ generated from our simulations, we can determine if the model is systematically failing to capture the global structure of the data.

4.2.2 Choice of predictive distribution

When generating replicated datasets, $y^{rep} = (y_1^{rep}, \dots, y_n^{rep})$, to assess model fit, we must decide which predictive distribution to sample from. Each choice stresses a different part of the Bayesian model.

1. Prior predictive, $p(\tilde{y}_i)$. This is used to assess the model before observing any data. Unless the prior is highly informative and accurate, the prior predictive distribution will typically generate synthetic data with far greater variability than the actual observed data.
2. Posterior predictive, $p(\tilde{y}_i|y)$. This is the most common choice. It generates replicates based on the parameters updated by the entire observed dataset.

This is sometimes criticized for double-using the data, as y_i is used both to train the model and to judge its performance. This can make the model look better than it actually is.

3. Cross validated posterior predictive, $p(\tilde{y}_i|y_{(-i)})$. Also known as the Leave-one-out predictive distribution, this generates a replicate for the i -th observation using a posterior conditioned on all data except y_i . It is more rigorous than the posterior predictive because it avoids the double-use problem.

Unless sample size is small or a specific observation y_i is highly influential, the replicates generated by this method are similar to those from the standard posterior predictive. Because it is computationally more demanding, the standard posterior predictive is often favored for general diagnostics.

4.2.3 Choice of discrepancy measure

Once a test statistic $T(y)$ has been chosen, we must measure the degree of conflict between the observed value $T(y)$ and its predictive distribution. There are two primary ways to quantify this discrepancy:

- a) The most common measure is the *tail-area* probability, which calculates how extreme the observed statistic is relative to the simulations. It is defined as:

$$\text{Probability}(T(\tilde{y}) \leq T(y)), \text{ or } \text{Probability}(T(\tilde{y}) \geq T(y)).$$

The choice depends on the nature of the statistic. We use the tail that corresponds to worse fit. The probability can be computed with respect to either one of the predictive distributions considered.

A large, but not too-large, tail-area probability indicates that the observed data is a typical realization of the model. Conversely, a value near 0 or 1 suggests the model is failing to capture that specific feature of the data.

This value is easily estimated by calculating the proportion of simulated replicates $T(y^{rep})$ that are more extreme than the observed $T(y)$.

- b) A less common but interesting alternative is the *relative surprise index*, defined as the ratio of the density at the observed value to the maximum possible density:

$$RSI = \frac{h(T(y))}{\max_{\tilde{y}} h(T(\tilde{y}))},$$

where $h(\cdot)$ is the pdf of the predictive distribution for $T(\tilde{y})$.

An RSI near 1 means the observed data lies at the most likely point of the predictive distribution.

Because RSI requires knowledge of the full density function rather than just samples, it is harder to estimate via simulation and is used less in practice.

Beyond numerical indices, graphical representations offer a powerful, intuitive check. A common technique is the line-up test. You mix a plot of the real data with several plots of synthetic data and check if an observer can distinguish the real from the replicated.

If the real data stands out, the visual differences provide immediate, qualitative hints about what the model is missing (e.g., it fails to capture clusters, it underestimates tails, or it misses a trend). This visual feedback is often the fastest path to model improvement.

Example 12.

See Puig, X., and Ginebra, J., (2014). “A Bayesian cluster analysis of election results,” *Journal of Applied Statistics*, 41, 73-94. <https://doi.org/10.1080/02664763.2013.830088>

Example 13.

See Puig, X., and Ginebra, J., (2023). “Mapping life expectancy loss in Barcelona in 2020,” *The American statistician*, 77, 417-424. <https://doi.org/10.1080/00031305.2023.2197022>

